Nonparametric Predictive Regression with Exogenous Shocks: A Sieve Empirical Likelihood Approach (In progress Version 2.0 AR1)

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1 Introduction and Data-Generating Process

We consider the problem of testing return predictability in the presence of highly persistent predictors and exogenous covariates (such as weather shocks). The data-generating process (DGP) is specified by the following system of equations:

1. **Predictive Regression:**

$$Y_t = m(W_{t-1}) + \beta X_{t-1} + U_t \tag{1}$$

$$X_t = \theta + \rho X_{t-1} + \varepsilon_t \tag{2}$$

2. **Endogeneity:**

$$U_t = \phi\varepsilon_t + z_t \quad \text{with } E[z_t \mid \varepsilon_t] = 0 \tag{3}$$

where Y_t represents the asset returns, W_{t-1} is a stationary exogenous weather covariate, and X_{t-1} is a persistent financial predictor that is independent of W_{t-1}. The error terms are structured to accommodate potential endogeneity via the parameter ϕ.

Motivated by Perron (1989), our goal is to investigate return predictability while avoiding size distortions or false inferences caused by near unit roots in the predictor dynamics (i.e., $\rho \approx 1$). This is achieved by testing the null hypothesis of no predictability:

$$\begin{aligned} H_0 &: \quad \beta = 0 \\ H_1 &: \quad \beta \neq 0 \end{aligned} \tag{4}$$

By extending the empirical likelihood (EL) framework of Cai et al. (2026), we propose a two-stage Orthogonalized Sieve-EL framework that achieves a standard χ_1^2 limiting distribution under the null.

2 Empirical Motivation: Weather Shocks and Cocoa Futures

We motivate our framework with the 2023–2024 agricultural commodity shocks. Let Y_t denote excess returns on cocoa futures, W_{t-1} an exogenous local weather variable such as precipitation or average temperature, and X_{t-1} a persistent financial predictor, such as the lagged price.

During 2023–2024, extreme weather conditions, including severe droughts in West Africa, led to supply deficits and a substantial increase in cocoa prices. This context illustrates the key features of our model:

1. **Exogeneity of Weather Shocks:** Local weather variables (W_{t-1}) are determined by atmospheric systems, exogenous to financial variables (X_{t-1}) such as trading volume or lagged prices. Thus, W_{t-1} and X_{t-1} are independent, satisfying Assumption A.

2. **Nonlinear Weather Impact:** The effect of temperature and rainfall on crop yield is highly nonlinear, with extreme weather deviations causing disproportionate impacts on supply. A nonparametric formulation $m(W_{t-1})$ is necessary to capture this threshold behavior, which we approximate using a sieve basis.

3. **Predictor Persistence:** Financial predictors (X_{t-1}) often exhibit near unit root behavior ($\rho \approx 1$). Standard OLS inference on β with $m(W_{t-1})$ and persistent X_{t-1} leads to size distortions.

4. **Orthogonalized Sieve-EL Application:** Our framework first filters the nonlinear weather impact $m(W_{t-1})$ via sieve projection. The independence of W_{t-1} and X_{t-1} ensures consistency. Centering the predictor's weights against the sieve basis purges the nonparametric estimation error from the EL moment condition, allowing for standard χ_1^2 inference on β.

3 The Orthogonalized Sieve-EL Framework (Version 2.0 AR1)

We handle the unknown nonparametric curve $m(W_{t-1})$ by approximating it using a sieve basis of dimension K (e.g., polynomials or splines). Let $\boldsymbol{P}_K(W_{t-1}) = (p_1(W_{t-1}), \ldots, p_K(W_{t-1}))'$ be the $K \times 1$ vector of these basis functions.

3.1 The Step-by-Step Algorithm

The estimation and inference procedure is implemented sequentially via the following five steps, detailing the explicit matrix algebra that guarantees robustness.

Step 1: The First-Stage AR(1) (Purging the Predictor)

Following the methodology of Cai & Wang (2014), we first isolate the true innovations of the highly persistent predictor X_t. We estimate a first-order autoregression via OLS:

$$X_t = \theta + \rho X_{t-1} + \varepsilon_t, \quad t = 1, \dots, T \tag{5}$$

The OLS estimators for the intercept and persistence parameter are given by:

$$\begin{pmatrix} \widehat{\theta} \\ \widehat{\rho} \end{pmatrix} = \left(\sum_{t=1}^{T} \begin{pmatrix} 1 \\ X_{t-1} \end{pmatrix} \begin{pmatrix} 1 & X_{t-1} \end{pmatrix} \right)^{-1} \sum_{t=1}^{T} \begin{pmatrix} 1 \\ X_{t-1} \end{pmatrix} X_t \tag{6}$$

We then compute and save the estimated residuals, which act as our proxy for the pure financial shock:

$$\widehat{\varepsilon}_t = X_t - \widehat{\theta} - \widehat{\rho} X_{t-1} \tag{7}$$

Step 2: The Joint Sieve-Cai-Wang Estimation (Purging the Weather Curve)

To estimate the nonparametric weather curve $m(W_{t-1})$ without absorbing any endogenous financial shocks, we set up a joint OLS regression. The return shock is decomposed as $U_t = \phi \varepsilon_t + z_t$. Substituting this into the return equation yields the full regression model:

$$Y_t = \boldsymbol{P}_K(W_{t-1})' \boldsymbol{c} + \beta X_{t-1} + \phi \widehat{\varepsilon}_t + \text{error}_t \tag{8}$$

Let $\boldsymbol{\Lambda}_t = (\boldsymbol{P}_K(W_{t-1})', X_{t-1}, \widehat{\varepsilon}_t)'$ denote the $(K+2) \times 1$ concatenated regressor vector. The joint OLS estimator $\widehat{\boldsymbol{\delta}} = \left(\widehat{\boldsymbol{c}}', \widehat{\beta}, \widehat{\phi} \right)'$ is defined as:

$$\widehat{\boldsymbol{\delta}} = \left(\sum_{t=1}^{T} \boldsymbol{\Lambda}_t \boldsymbol{\Lambda}_t' \right)^{-1} \sum_{t=1}^{T} \boldsymbol{\Lambda}_t Y_t \tag{9}$$

We extract the first K elements of $\widehat{\boldsymbol{\delta}}$ to serve as our robust estimator for the Sieve coefficients, $\widehat{\boldsymbol{c}}$.

- **No Omitted Variable Bias:** By explicitly including X_{t-1} in $\boldsymbol{\Lambda}_t$, the projection of Y_t onto the Sieve basis is orthogonalized against the predictor, preventing $\widehat{\boldsymbol{c}}$ from spuriously absorbing the predictive signal $\beta_0 X_{t-1}$.
- **No Stambaugh Bias:** By including $\widehat{\varepsilon}_t$, the endogeneity correlation $\mathbb{E}[U_t \varepsilon_t] \neq 0$ is explicitly partialled out. The remaining error is strictly orthogonal to the predictor's innovations, thereby unbiasing the estimation.

Step 3: Redefining the Adjusted Return

With a clean estimator $\widehat{\boldsymbol{c}}$ in hand, we isolate the financial components of the return by subtracting the estimated nonparametric weather impact. The adjusted dependent variable is defined as:

$$\widetilde{Y}_t = Y_t - \boldsymbol{P}_K(W_{t-1})'\widehat{\boldsymbol{c}} \tag{10}$$

Under the null hypothesis $H_0 : \beta = \beta_0$, substituting the true DGP $Y_t = m(W_{t-1}) + \beta_0 X_{t-1} + U_t$ and the Sieve decomposition $m(W_{t-1}) = \boldsymbol{P}_K(W_{t-1})'\boldsymbol{c} + r_t$ (where r_t is the approximation error) yields:

$$\begin{aligned}
\widetilde{Y}_t - \beta_0 X_{t-1} &= m(W_{t-1}) + U_t - \boldsymbol{P}_K(W_{t-1})'\widehat{\boldsymbol{c}} && (11)\\
&= \boldsymbol{P}_K(W_{t-1})'\boldsymbol{c} + r_t + U_t - \boldsymbol{P}_K(W_{t-1})'\widehat{\boldsymbol{c}} && (12)\\
&= U_t + r_t - \boldsymbol{P}_K(W_{t-1})'(\widehat{\boldsymbol{c}} - \boldsymbol{c}) && (13)
\end{aligned}$$

Step 4: The Sieve Weight Projection

In standard Empirical Likelihood, the estimation error $(\widehat{\boldsymbol{c}} - \boldsymbol{c})$ would severely distort the asymptotic distribution. To neutralize this, we construct an orthogonalized weight function. Let $w(X_{t-1})$ be the raw weight (e.g., $\tanh(X_{t-1}/c_w)$). We project $w(X_{t-1})$ onto the Sieve basis:

$$\widehat{\boldsymbol{\gamma}} = \left(\sum_{t=1}^{T} \boldsymbol{P}_K(W_{t-1})\boldsymbol{P}_K(W_{t-1})'\right)^{-1} \sum_{t=1}^{T} \boldsymbol{P}_K(W_{t-1})w(X_{t-1}) \tag{14}$$

The orthogonalized, centered weight w_t^c is defined as the residual of this regression:

$$w_t^c = w(X_{t-1}) - \boldsymbol{P}_K(W_{t-1})'\widehat{\boldsymbol{\gamma}} \tag{15}$$

By the fundamental first-order conditions of OLS, the residuals are exactly orthogonal to the regressors in-sample:

$$\sum_{t=1}^{T} \boldsymbol{P}_K(W_{t-1})w_t^c \equiv \boldsymbol{0} \tag{16}$$

Step 5: Algebraic Elimination in the EL Moment Condition

For testing $H_0 : \beta = \beta_0$, the EL moment condition is evaluated as:

$$Z_t(\beta_0) = \left(\widetilde{Y}_t - \beta_0 X_{t-1}\right) w_t^c \tag{17}$$

Substituting the expansion from Step 3:

$$Z_t(\beta_0) = \Big(U_t + r_t - \boldsymbol{P}_K(W_{t-1})'(\widehat{\boldsymbol{c}} - \boldsymbol{c})\Big)w_t^c \tag{18}$$

Summing the moment conditions across the sample to evaluate the numerator of the EL statistic:

$$\sum_{t=1}^{T} Z_t(\beta_0) = \sum_{t=1}^{T}(U_t + r_t)w_t^c - (\widehat{\boldsymbol{c}} - \boldsymbol{c})' \underbrace{\sum_{t=1}^{T} \boldsymbol{P}_K(W_{t-1})w_t^c}_{= \mathbf{0} \text{ (from Step 4)}} \tag{19}$$

The multi-dimensional Sieve estimation error $\widehat{\boldsymbol{c}} - \boldsymbol{c}$ is thereby perfectly annihilated algebraically. We are left with:

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} Z_t(\beta_0) = \frac{1}{\sqrt{T}} \sum_{t=1}^{T} U_t w_t^c + \underbrace{\frac{1}{\sqrt{T}} \sum_{t=1}^{T} r_t w_t^c}_{= o_p(1)} \tag{20}$$

Because the condition is structurally stripped of estimation variance, the sample variance of $Z_t(\beta_0)$ accurately reflects the pure innovation variance. As a result, the Empirical Likelihood ratio statistic (where $\widehat{\lambda}$ is the Lagrange multiplier) achieves the standard chi-squared limit:

$$\ell_T(\beta_0) = 2 \sum_{t=1}^{T} \log(1 + \widehat{\lambda} Z_t(\beta_0)) \xrightarrow{d} \chi_1^2 \tag{21}$$

3.2 Comparison with Cai et al. (2026)

Our framework differs from the two-stage approach of Cai et al. (2026) in three fundamental aspects:

- **Nuisance Dimensionality**: Cai et al. (2026) handle a fixed scalar intercept α, whereas our model contains an infinite-dimensional nuisance function $m(W_{t-1})$, which requires a sieve approximation of dimension $K \to \infty$.
- **Orthogonalization Space**: In Cai et al. (2026), the weight centering $w^c(x) = w(x) - \frac{1}{T}\sum w(X_s)$ represents a projection onto the constant subspace span$\{\mathbf{1}\}$. In contrast, our orthogonalized weight w_t^c is projected onto the K-dimensional sieve space span$\{\boldsymbol{P}_K(W_{t-1})\}$.
- **Purging and Exogeneity**: By jointly estimating the sieve basis alongside X_{t-1} and $\widehat{\varepsilon}_t$, we ensure the nonparametric components are fully immune to both omitted variable bias and Stambaugh endogeneity bias, mirroring the core philosophy of CCHT.

4 The Growth Rate of the Sieve Dimension K

To establish the asymptotic validity of our test statistic, we must place bounds on the growth rate of the tuning parameter K relative to the sample size T.

4.1 Lower Bound (Bias Elimination)

Assume the true weather curve $m(W)$ has smoothness s (i.e., is s times continuously differentiable). Sieve theory implies that the approximation error r_t satisfies $\sup |r_t| = O_p(K^{-s})$. To ensure that the approximation bias does not distort the asymptotic distribution, we require $\sqrt{T} \sum_{t=1}^{T} r_t w_t^c = o_p(1)$, which translates to:

$$\sqrt{T} K^{-s} \to 0 \quad \Longrightarrow \quad K \gg T^{\frac{1}{2s}} \tag{22}$$

4.2 Upper Bound (Variance Control and Leakage Mitigation)

Estimating K parameters in OLS introduces in-sample variance inflation and potential "future leakage" bias. Although the α-mixing property of the weather process W_t with exponential decay prevents leakage from accumulating, the pure variance penalty scales as $K/\sqrt{T}$. To prevent variance inflation from inflating the denominator of the EL ratio, we require:

$$\frac{K}{\sqrt{T}} \to 0 \quad \Longrightarrow \quad K \ll T^{\frac{1}{2}} \tag{23}$$

4.3 Growth rate of K

Combining these bounds yields the formal asymptotic growth condition:

$$T^{\frac{1}{2s}} \ll K \ll T^{\frac{1}{2}} \tag{24}$$

If we assume $m(W)$ is twice-differentiable ($s = 2$), the condition becomes $T^{1/4} \ll K \ll T^{1/2}$. In practice, setting $K \approx T^{1/3}$ satisfies these bounds, ensuring that both approximation bias and variance inflation vanish asymptotically.

5 The Formal Independence Assumption

To ensure the algebraic orthogonalization rigorously eliminates the nonparametric estimation bias, we must formally state the economic relationship between the financial market predictor and the exogenous weather shock. We impose the following two independence conditions on W_{t-1}:

Assumption 5.1 (Strict Exogeneity to the Market Shock) *The weather variable W_{t-1} is strictly exogenous to the financial market innovation U_t. Mathematically:*

$$\mathbb{E}[U_t \mid \mathcal{F}_{t-1}, W_{t-1}, W_t, W_{t+1}, \ldots] = 0. \tag{25}$$

Remark. This condition ensures that using the entire dataset to estimate the weather curve does not inadvertently absorb future stock market shocks. It shuts down the "future leakage" trap. Because climatic processes are governed by atmospheric physics, weather shocks are completely oblivious to tomorrow's stock market crashes.

Assumption 5.2 (Cross-Sectional Independence from the Predictor) *The exogenous weather process $\{W_t\}$ is strictly independent of the financial predictor innovation process $\{v_t\}$. Consequently, W_{t-1} and X_{t-1} are independent random variables, implying:*

$$\mathbb{E}[w(X_{t-1}) \mid W_{t-1}] = \mathbb{E}[w(X_{t-1})]. \tag{26}$$

Remark. This assumption fundamentally preserves the predictor's signal. Because W_{t-1} and X_{t-1} are independent, projecting the EL weight $w(X_{t-1})$ onto the weather basis $\boldsymbol{P}_K(W_{t-1})$ produces an R^2 of exactly zero asymptotically. The sieve projection structurally deletes the nonparametric estimation bias without extracting a single drop of the persistent signal from X_{t-1}.

6 Wilks' Theorem for Mean Predictability

This section gives a formal Wilks theorem for the mean-predictability test. It is useful to write the nuisance step in null-imposed profile form. Let

$$\boldsymbol{Y} = (Y_1, \ldots, Y_T)', \qquad \boldsymbol{X}_- = (X_0, \ldots, X_{T-1})',$$

and let $\boldsymbol{P}$ be the $T \times K$ matrix whose t-th row is $\boldsymbol{P}_K(W_{t-1})'$. Define

$$\boldsymbol{\Pi}_K = \boldsymbol{P}(\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}', \qquad \boldsymbol{M}_K = \boldsymbol{I}_T - \boldsymbol{\Pi}_K.$$

For each candidate value β, define the null-imposed profile sieve coefficient and the corresponding residual by

$$\widehat{\boldsymbol{c}}_K(\beta) = (\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'(\boldsymbol{Y} - \beta\boldsymbol{X}_-), \qquad \widehat{\boldsymbol{u}}(\beta) = \boldsymbol{M}_K(\boldsymbol{Y} - \beta\boldsymbol{X}_-). \tag{27}$$

Let $w_t = w(X_{t-1})$, $\boldsymbol{w} = (w_1, \ldots, w_T)'$, and define the sieve-orthogonalized weight

$$\boldsymbol{w}^c = \boldsymbol{M}_K\boldsymbol{w}, \qquad w_t^c = (\boldsymbol{w}^c)_t. \tag{28}$$

The scalar estimating equation is

$$\widehat{Z}_t(\beta) = \widehat{u}_t(\beta)w_t^c. \tag{29}$$

The empirical likelihood and its log-likelihood ratio are

$$L_T(\beta) = \sup\left\{\prod_{t=1}^{T}(T\pi_t) : \pi_t \geq 0, \quad \sum_{t=1}^{T}\pi_t = 1, \quad \sum_{t=1}^{T}\pi_t\widehat{Z}_t(\beta) = 0\right\}, \tag{30}$$

$$\ell_T(\beta) = -2\log L_T(\beta) = 2\sum_{t=1}^{T}\log\{1 + \widehat{\lambda}(\beta)\widehat{Z}_t(\beta)\}, \tag{31}$$

where $\widehat{\lambda}(\beta)$ solves

$$\sum_{t=1}^{T} \frac{\widehat{Z}_t(\beta)}{1+\widehat{\lambda}(\beta)\widehat{Z}_t(\beta)} = 0. \tag{32}$$

For a vector $\boldsymbol{a} = (a_1, \ldots, a_T)'$, write $\|\boldsymbol{a}\|_T^2 = T^{-1}\sum_{t=1}^{T} a_t^2$. Also write

$$\overline{w}_T = T^{-1}\sum_{t=1}^{T} w_t, \qquad \widetilde{w}_t = w_t - \overline{w}_T.$$

Assumption 1 (CCHT mean-score conditions)**.** *The predictor satisfies $X_t = \theta + \rho_T X_{t-1} + \varepsilon_t$, where ρ_T belongs to one of the persistence classes considered by Cai, Chen, Hong, and Tsvetanov (CCHT), including the stationary, mildly integrated, local-to-unity, integrated, and mildly explosive cases. With respect to the filtration*

$$\mathcal{H}_t = \sigma\{(z_s, \varepsilon_s) : s \le t\} \vee \sigma\{W_s : s \in \mathbb{Z}\},$$

$E(U_t \mid \mathcal{H}_{t-1}) = 0$, $\sup_t E|U_t|^{2+\delta} < \infty$ for some $\delta > 0$, and the conditional-variance and weight conditions used for the CCHT mean theorem hold. In particular, for

$$A_T = \frac{1}{\sqrt{T}}\sum_{t=1}^{T} U_t \widetilde{w}_t, \qquad V_T = \frac{1}{T}\sum_{t=1}^{T} U_t^2 \widetilde{w}_t^2,$$

there exists an a.s. positive, possibly random, variable η^2 such that

$$A_T \xrightarrow{\text{st}} MN(0, \eta^2), \qquad V_T \xrightarrow{p} \eta^2, \qquad \max_{1\le t\le T} |U_t \widetilde{w}_t| = o_p(\sqrt{T}). \tag{33}$$

The raw weight is uniformly bounded and satisfies the CCHT saturation condition.

Assumption 2 (Weather process and sieve projection)**.** *The entire weather process is independent of $\{X_0, z_t, \varepsilon_t : t \ge 1\}$, and the strict-exogeneity condition in Assumption 1 holds after enlarging the filtration by the entire weather path. The first component of $\boldsymbol{P}_K(W)$ is one. After centering the remaining basis functions in sample, write the corresponding $T \times (K-1)$ matrix as $\boldsymbol{R}$, so that $\boldsymbol{1}_T'\boldsymbol{R} = \boldsymbol{0}$ and $\operatorname{span}(\boldsymbol{P}) = \operatorname{span}(\boldsymbol{1}_T, \boldsymbol{R})$. There are constants $0 < c < C < \infty$ such that, with probability tending to one,*

$$c \le \lambda_{\min}(T^{-1}\boldsymbol{R}'\boldsymbol{R}) \le \lambda_{\max}(T^{-1}\boldsymbol{R}'\boldsymbol{R}) \le C.$$

Let $\zeta_K = \max_{t\le T}\|\boldsymbol{R}_t\|$, where $\boldsymbol{R}_t'$ is row t of $\boldsymbol{R}$. The following increasing-dimensional score bounds hold:

$$\left\|T^{-1}\boldsymbol{R}'\widetilde{\boldsymbol{w}}\right\| = O_p\left(\sqrt{K/T}\right), \tag{34}$$

$$\left\|T^{-1/2}\boldsymbol{R}'\boldsymbol{U}\right\| = O_p(\sqrt{K}), \tag{35}$$

$$\left\|T^{-1}\boldsymbol{P}'\boldsymbol{U}\right\| = O_p\left(\sqrt{K/T}\right), \tag{36}$$

where $\widetilde{\boldsymbol{w}} = (\widetilde{w}_1, \ldots, \widetilde{w}_T)'$ *and* $\boldsymbol{U} = (U_1, \ldots, U_T)'$. *These bounds are implied, for example, by process independence, exponential mixing of* W_t, *uniformly bounded basis moments, and the martingale moment condition in Assumption 1.*

Assumption 3 (Approximation and growth rates)**.** *For some* $\boldsymbol{c}_K$,

$$m(W_{t-1}) = \boldsymbol{P}_K(W_{t-1})'\boldsymbol{c}_K + r_{K,t},$$

where

$$\|\boldsymbol{r}_K\|_T = O(a_K), \qquad \|\boldsymbol{r}_K\|_\infty = O(a_K), \qquad \sqrt{T}\, a_K \longrightarrow 0.$$

The sieve projector is stable in sup norm on the approximation residual, so that $\|\boldsymbol{M}_K \boldsymbol{r}_K\|_\infty = O(a_K)$. *Moreover,*

$$K/\sqrt{T} \longrightarrow 0, \qquad \zeta_K \sqrt{K/T} \longrightarrow 0. \tag{37}$$

For normalized spline bases, for which typically $\zeta_K \asymp \sqrt{K}$, *the second condition in* (37) *is implied by* $K/\sqrt{T} \to 0$. *If* $a_K \asymp K^{-s/d}$, *a sufficient rate window is* $T^{d/(2s)} \ll K \ll T^{1/2}$, *which is nonempty when* $s > d$.

Assumption 4 (Nondegeneracy and convex hull)**.** *There exists* $\underline{\eta} > 0$ *such that* $P(\eta^2 \geq \underline{\eta}^2) = 1$. *In addition,*

$$P\left\{\min_{1\leq t\leq T} \widehat{Z}_t(\beta_0) < 0 < \max_{1\leq t\leq T} \widehat{Z}_t(\beta_0)\right\} \longrightarrow 1.$$

Lemma 1 (Negligibility of the growing sieve projection)**.** *Under Assumptions 2–3,*

$$\|\boldsymbol{w}^c - \widetilde{\boldsymbol{w}}\|_T = O_p\left(\sqrt{K/T}\right), \tag{38}$$

$$\max_{t\leq T} |w_t^c - \widetilde{w}_t| = O_p\left(\zeta_K \sqrt{K/T}\right) = o_p(1), \tag{39}$$

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} U_t(w_t^c - \widetilde{w}_t) = O_p(K/\sqrt{T}) = o_p(1), \tag{40}$$

$$\|\boldsymbol{\Pi}_K \boldsymbol{U}\|_T = O_p\left(\sqrt{K/T}\right), \tag{41}$$

$$\max_{t\leq T} |(\boldsymbol{\Pi}_K \boldsymbol{U})_t| = O_p\left(\zeta_K \sqrt{K/T}\right) = o_p(1). \tag{42}$$

Proof. Because the sieve contains a constant and $\boldsymbol{1}_T'\boldsymbol{R} = \boldsymbol{0}$, residualizing $\boldsymbol{w}$ on $\boldsymbol{P}$ is equivalent to residualizing $\widetilde{\boldsymbol{w}}$ on $\boldsymbol{R}$. Hence

$$\boldsymbol{w}^c = \widetilde{\boldsymbol{w}} - \boldsymbol{R}\widehat{\boldsymbol{\delta}}_K, \qquad \widehat{\boldsymbol{\delta}}_K = (\boldsymbol{R}'\boldsymbol{R})^{-1}\boldsymbol{R}'\widetilde{\boldsymbol{w}}.$$

The Gram-matrix condition and (34) imply $\|\widehat{\boldsymbol{\delta}}_K\| = O_p(\sqrt{K/T})$. Therefore,

$$\|\boldsymbol{R}\widehat{\boldsymbol{\delta}}_K\|_T^2 = \widehat{\boldsymbol{\delta}}_K'(T^{-1}\boldsymbol{R}'\boldsymbol{R})\widehat{\boldsymbol{\delta}}_K = O_p(K/T),$$

which proves (38); and

$$\max_t |\boldsymbol{R}_t'\widehat{\boldsymbol{\delta}}_K| \le \zeta_K \|\widehat{\boldsymbol{\delta}}_K\| = O_p\left(\zeta_K\sqrt{K/T}\right),$$

which proves (39). Moreover,

$$\frac{1}{\sqrt{T}}\boldsymbol{U}'(\boldsymbol{w}^c - \widetilde{\boldsymbol{w}}) = -\left(T^{-1/2}\boldsymbol{R}'\boldsymbol{U}\right)'\widehat{\boldsymbol{\delta}}_K = O_p(\sqrt{K})O_p\left(\sqrt{K/T}\right) = O_p(K/\sqrt{T}),$$

giving (40).

Finally, let $\widehat{\boldsymbol{a}}_U = (\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'\boldsymbol{U}$. The Gram condition and (36) imply $\|\widehat{\boldsymbol{a}}_U\| = O_p(\sqrt{K/T})$. The same quadratic-form and maximum-row arguments give (41) and (42). $\square$

Lemma 2 (Asymptotic equivalence of the feasible and oracle scores)**.** *Under Assumptions 1–3, under $H_0 : \beta = \beta_0$,*

$$S_T := \frac{1}{\sqrt{T}}\sum_{t=1}^{T}\widehat{Z}_t(\beta_0) = \frac{1}{\sqrt{T}}\sum_{t=1}^{T}U_t\widetilde{w}_t + o_p(1), \tag{43}$$

$$\widehat{V}_T := \frac{1}{T}\sum_{t=1}^{T}\widehat{Z}_t(\beta_0)^2 = \frac{1}{T}\sum_{t=1}^{T}U_t^2\widetilde{w}_t^2 + o_p(1), \tag{44}$$

$$\max_{t\le T}|\widehat{Z}_t(\beta_0)| = o_p(\sqrt{T}). \tag{45}$$

Proof. Under the null,

$$\boldsymbol{Y} - \beta_0\boldsymbol{X}_- = \boldsymbol{P}\boldsymbol{c}_K + \boldsymbol{r}_K + \boldsymbol{U}, \qquad \widehat{\boldsymbol{u}}(\beta_0) = \boldsymbol{M}_K(\boldsymbol{U} + \boldsymbol{r}_K).$$

Because $\boldsymbol{M}_K$ is symmetric and idempotent and $\boldsymbol{w}^c = \boldsymbol{M}_K\boldsymbol{w}$,

$$\begin{aligned} S_T &= T^{-1/2}\widehat{\boldsymbol{u}}(\beta_0)'\boldsymbol{w}^c \\ &= T^{-1/2}(\boldsymbol{U} + \boldsymbol{r}_K)'\boldsymbol{M}_K^2\boldsymbol{w} \\ &= T^{-1/2}(\boldsymbol{U} + \boldsymbol{r}_K)'\boldsymbol{w}^c. \end{aligned}$$

Lemma 1 gives $T^{-1/2}\boldsymbol{U}'\boldsymbol{w}^c = T^{-1/2}\boldsymbol{U}'\widetilde{\boldsymbol{w}} + o_p(1)$. Also, by Cauchy–Schwarz and the contraction property of an orthogonal projection,

$$\left|T^{-1/2}\boldsymbol{r}_K'\boldsymbol{w}^c\right| \le \sqrt{T}\,\|\boldsymbol{r}_K\|_T\|\boldsymbol{w}^c\|_T \le \sqrt{T}\,\|\boldsymbol{r}_K\|_T\|\boldsymbol{w}\|_T = o_p(1).$$

This proves (43).

For the second-moment result, define

$$e_{u,t} = \widehat{u}_t(\beta_0) - U_t, \qquad e_{w,t} = w_t^c - \widetilde{w}_t.$$

Then

$$\boldsymbol{e}_u = -\boldsymbol{\Pi}_K\boldsymbol{U} + \boldsymbol{M}_K\boldsymbol{r}_K,$$

so that, by Lemma 1 and projection contraction,

$$\|\boldsymbol{e}_u\|_T = O_p\Big(\sqrt{K/T} + a_K\Big) = o_p(1).$$

Moreover, $\max_t |e_{w,t}| = o_p(1)$, $\max_t |w_t^c| = O_p(1)$, and $T^{-1}\sum_t U_t^2 = O_p(1)$. Since

$$\widehat{Z}_t(\beta_0) - U_t \widetilde{w}_t = e_{u,t} w_t^c + U_t e_{w,t},$$

it follows that

$$\begin{aligned}\frac{1}{T}\sum_{t=1}^{T}\{\widehat{Z}_t(\beta_0) - U_t\widetilde{w}_t\}^2 &\le 2\max_t |w_t^c|^2 \|\boldsymbol{e}_u\|_T^2 + 2\max_t |e_{w,t}|^2 \frac{1}{T}\sum_{t=1}^{T} U_t^2 \\ &= o_p(1).\end{aligned}$$

Equation (44) now follows from Cauchy–Schwarz.

Finally, $\max_t |U_t| = o_p(\sqrt{T})$ follows from the uniform $(2+\delta)$-moment bound. Together with (42), the sup-norm stability of $\boldsymbol{M}_K \boldsymbol{r}_K$, and $\max_t |w_t^c| = O_p(1)$, this proves (45). $\square$

Theorem 1 (Wilks phenomenon for sieve-orthogonalized mean EL)**.** *Suppose Assumptions 1–4 hold. Then, under $H_0 : \beta = \beta_0$,*

$$\ell_T(\beta_0) \xrightarrow{d} \chi_1^2. \tag{46}$$

The conclusion holds for each predictor-persistence class covered by the CCHT mean theorem.

Proof. By Lemma 2 and Assumption 1,

$$S_T \xrightarrow{\text{st}} MN(0, \eta^2), \qquad \widehat{V}_T \xrightarrow{p} \eta^2.$$

Consequently,

$$\frac{S_T}{\sqrt{\widehat{V}_T}} \xrightarrow{\text{st}} N(0,1), \qquad \frac{S_T^2}{\widehat{V}_T} \xrightarrow{d} \chi_1^2. \tag{47}$$

It remains to connect the self-normalized statistic to empirical likelihood. Write $\widehat{Z}_t = \widehat{Z}_t(\beta_0)$ and $\widehat{\lambda} = \widehat{\lambda}(\beta_0)$. The convex-hull condition, $S_T = O_p(1)$, $\widehat{V}_T \to_p \eta^2 > 0$, and $\max_t |\widehat{Z}_t| = o_p(\sqrt{T})$ imply the standard scalar-EL bounds

$$\widehat{\lambda} = O_p(T^{-1/2}), \qquad \max_{t \le T} |\widehat{\lambda}\widehat{Z}_t| = o_p(1).$$

Expanding the multiplier equation (32) gives

$$0 = \sum_{t=1}^{T} \widehat{Z}_t - \widehat{\lambda}\sum_{t=1}^{T} \widehat{Z}_t^2 + o_p(\sqrt{T}),$$

and hence

$$\widehat{\lambda} = \frac{\sum_{t=1}^{T} \widehat{Z}_t}{\sum_{t=1}^{T} \widehat{Z}_t^2} + o_p(T^{-1/2}). \tag{48}$$

A second-order expansion of (31), together with $\max_t |\widehat{\lambda}\widehat{Z}_t| = o_p(1)$, yields

$$\begin{aligned}
\ell_T(\beta_0) &= 2\widehat{\lambda} \sum_{t=1}^{T} \widehat{Z}_t - \widehat{\lambda}^2 \sum_{t=1}^{T} \widehat{Z}_t^2 + o_p(1) \\
&= \frac{\left(T^{-1/2} \sum_{t=1}^{T} \widehat{Z}_t\right)^2}{T^{-1} \sum_{t=1}^{T} \widehat{Z}_t^2} + o_p(1) \\
&= \frac{S_T^2}{\widehat{V}_T} + o_p(1).
\end{aligned}$$

The result follows from (47) and Slutsky's theorem. □

Corollary 1 (Test of no mean predictability)**.** *For $H_0 : \beta = 0$, reject at asymptotic level α whenever*

$$\ell_T(0) > \chi^2_{1,1-\alpha}.$$

Equivalently, the asymptotic p-value is $1 - F_{\chi^2_1}\{\ell_T(0)\}$. A confidence set for β is obtained by inversion:

$$\mathcal{C}_{1-\alpha} = \{\beta : \ell_T(\beta) \leq \chi^2_{1,1-\alpha}\}.$$

Remark 1 (Relation to the unrestricted two-stage estimator)**.** *The profile construction (27) is the cleanest route to Theorem 1: at β_0, the fitted nuisance error is exactly $\mathbf{\Pi}_K(\boldsymbol{U} + \boldsymbol{r}_K)$, whose prediction norm is $O_p(\sqrt{K/T} + a_K)$. If one instead retains a separately estimated unrestricted coefficient $\widehat{\boldsymbol{c}}$, the same theorem continues to hold provided one proves the additional generated-nuisance condition*

$$\frac{1}{T} \sum_{t=1}^{T} \left\{\boldsymbol{P}_K(W_{t-1})'(\widehat{\boldsymbol{c}} - \boldsymbol{c}_K)\right\}^2 (w_t^c)^2 = o_p(1)$$

and the corresponding maximum-term bound. Those statements do not follow from the algebraic identity $\boldsymbol{P}'\boldsymbol{w}^c = \mathbf{0}$ alone.

Remark 2 (Why the same argument is not immediately a quantile theorem)**.** *For a quantile score $\psi_\tau(u) = \tau - \mathbf{1}\{u < 0\}$, the nuisance error enters nonlinearly:*

$$\psi_\tau(U_{t,\tau} + r_{K,t} - \boldsymbol{P}_K(W_{t-1})'(\widehat{\boldsymbol{c}}_\tau - \boldsymbol{c}_{K,\tau}))\,.$$

Therefore $\boldsymbol{P}'\boldsymbol{w}^c = \mathbf{0}$ does not algebraically cancel the first-stage error. A first-order expansion instead produces a density-weighted term of the form

$$\sum_{t=1}^{T} f_{t,\tau}(0)\boldsymbol{P}_K(W_{t-1})w_t^c,$$

which is not generally zero. A quantile extension would require either a restrictive condition making $f_{t,\tau}(0)$ constant with respect to the predictor, or a new density-weighted orthogonalization and additional nonparametric-rate arguments. Thus Theorem 1 is a complete and natural endpoint for the mean paper, whereas a general quantile extension is a separate methodological problem.

7 Conclusion

When researchers use standard predictive regressions, they are often forced to assume a constant intercept to maintain mathematical tractability. Consequently, all other exogenous macro-environmental factors are relegated to the error term. While this preserves the null hypothesis, it artificially inflates the residual variance, mathematically obliterating the test's power to detect true financial predictability. Our Orthogonalized Sieve-EL estimator, augmented with the CCHT first-stage AR(1) purging step, allows researchers to surgically remove arbitrary, non-linear environmental noise without violating the stringent requirements of Empirical Likelihood or suffering from Stambaugh bias. By doing so, the framework fully recovers the lost signal of the persistent predictor, offering a powerful tool for complex, real-world scenarios such as commodity pricing.